

Lab 4 Activity

1. Load in the C02 data from the `datasets` package by running the following code:

```
data(C02, package = "datasets")
help(C02)
```

Look at the documentation to understand the variables in this dataset. Identify a way to analyze these data using a t test (several possible choices, but `conc` probably won't be an interesting DV). Evaluate each assumption, conduct the t test, compute an effect size, and interpret the results.

2. Load in the `sleep` data from the `datasets` package by running the following code:

```
data(sleep, package = "datasets")

# This makes analyzing the data easier

sleep <- sleep %>%
  tidyr::pivot_wider(id_cols = ID,
                     names_from = group,
                     names_prefix = "group",
                     values_from = extra)
```

Look at the documentation to understand the variables in this dataset (but try not to peek at the examples section of the documentation). Identify a way to analyze these data using a t test. Evaluate each assumption, conduct the t test, compute an effect size, and interpret the results.

3. The `t.test()` function also includes an `alternative =` argument. What does that argument do? How do your results differ if you try different values for the `alternative =` argument? Make sure to read the help menu of the `t.test()` function.
4. For (1) and (2), use `ggplot` to create interesting graphs to illustrate your results (what might you include in a thesis or published paper?).